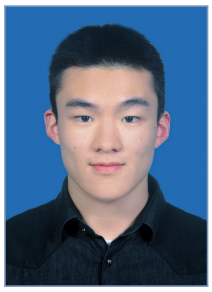
Junyi Hu

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(+86) 181-5591-2005 (same as WeChat) **Applying to PhD programs for Fall 2027**



The question that organizes my work is how visual systems can understand **interaction in the physical world**: how people, agents, objects and environments influence one another over time. I approach it from two connected directions, **visual navigation and tracking** and **human motion**, which meet when a robot learns from a person. In the long term, I want to make interaction an organizing principle for visual understanding.

EDUCATION

Fudan University B.S., School of Mathematical Sciences Sep. 2023 – Jun. 2027

GPA 3.60/4.00 | First-Class Undergraduate Scholarship (top 5%) | Courses: Differentiable Manifolds, Several Complex Variables, Cybernetics

PUBLICATIONS

1 **Learning Adaptive Safety Margins for Visual Navigation**
Junyi Hu, Shuaihang Yuan, Geeta Chandra Raju Bethala, Anthony Tzes, Yi Fang **IROS 2026** · First author
Made a navigation policy's safety margin a property of the situation: the robot learns from what it sees how much room to leave, more in clutter and less in open space, instead of following one fixed notion of safety that is too cautious in one place and too tight in another.

2 **VTaMo: Video–Text Alignment Model for Sign Language Translation**
Junyi Hu, Zhewen He, Haomian Huang, Aoxiang Yang, Yi Fang **ECCV 2026** · First author
Achieved unprecedented alignment between human motion and language, down to which stretch of movement carries which word, so that a continuous stream of motion can be read as the sequence of words it expresses, without ever being told where any word begins.

3 **SignNet-1M: Large-Scale Multilingual Sign Language Video Dataset with Downstream Benchmarks**
Zhewen He, **Junyi Hu**, Haomian Huang, Zhenhua Li, Yu-Shen Liu, Yi Fang **ECCV 2026** · Co-author
Built a million-clip video dataset of human motion whose viewpoint, scene and performer can be varied under control, factor by factor.

MANUSCRIPTS UNDER REVIEW

4 **CST-WM: A Causally Structured World Model for Embodied Visual Tracking**
Junyi Hu, Shuaihang Yuan, Jiazhao Liang, Yi Fang **Under review** · First author
Gave a world model for embodied tracking the causal structure of observer, target and environment, so the robot no longer believes that its own movement is what moves the person it follows, and can still plan how to find that person again whenever it loses sight of them.

5 **Keep the Effect, Drop the Actor: Programmable Effect-to-Execution World-Action Models**
Junyi Hu, Zhewen He, Yi Fang **Under review** · First author
Taught robots to read a human demonstration as a statement of how the world should change rather than as a motion to copy, so that a single video of a person becomes a task program that people can edit by hand and that robots with different bodies can then carry out.

6 **Real-Time Sign Language Translation by Prototype-Assisted Causal Streaming**
Junyi Hu, Zhewen He, Haomian Huang, Zhenhua Li, Yi Fang **Under review** · First author
Built the first real-time translation pipeline for the real world, fast and accurate: it turns live, noisy human motion into text as the motion unfolds, trusts each body joint only as far as its detector does, and commits each word once, as soon as it is sure, never taking it back.

7 **DINO Eats Geometry: Self-Supervised Sign Representations that Absorb Articulated Hand Geometry**
Junyi Hu, Zhewen He, Haomian Huang, Zhenhua Li, Zhifei Li, Shuaihang Yuan, Yi Fang **Under review** · First author
Learned representations of human motion that absorb the hand's articulated geometry during pre-training, so that the encoder knows how the fingers are configured and how they move at every frame, yet needs nothing but the video, and no hand model, at inference.

8 **HeatAtlas: Surface Textures as Learnable Atlases of Local Heat Kernels**
Zhewen He, **Junyi Hu**, Yi Fang **Under review** · Co-author
Made heat diffusion a local, learnable building block for appearance on 3D surfaces, with no global computation over the whole shape.

RESEARCH EXPERIENCE

New York University Abu Dhabi Undergraduate Researcher, advised by Prof. Yi Fang Jul. 2025 – Present
Led the projects above end to end, from problem formulation to deployment on real robots: a Unitree G1, a Unitree Go2 and a Franka FR3.

Liqing Intelligence Embodied AI Intern, on site Aug. 2026 – Present
Research and development on the embodied-intelligence stack of a world-model startup, together with market research across the field.

Intervertebral Disc Degeneration Grading from MRI NSFC Undergraduate Research Project Jul. 2026 – Present
Radiological pattern recognition and automated grading of disc degeneration from MRI, working with Shanghai Ninth People's Hospital.